

PROJECT SYNOPSIS

Create EC2 Instances in Public and Private Subnets and Test Connectivity

Project No. 6 · Computer Science & Engineering (CSE) — Cloud & DevOps

Branch / Stream	Computer Science & Engineering (CSE) — Cloud & DevOps
Group No.	Group 6
Group Leader	Prince (Roll No. 2415001181)
AWS Account ID	638596943163

Group Members

#	Name	Roll No.	Role in the Project
1	Prince	2415001181	Group Leader — EC2 instance deployment (public & private) and overall coordination
2	Sohit Sharma	2415001582	Phase 1 — VPC creation and public/private subnet setup
3	Mudit Chaudhary	2415000995	Phase 2 — Internet Gateway, route tables, and security group configuration
4	Anshika Rai	2415000261	Phase 4 — Connectivity testing and verification (SSH & ping)

Introduction

Amazon Web Services (AWS) allows resources to be organized inside a Virtual Private Cloud (VPC), a logically isolated network that can be divided into subnets with different levels of internet exposure. This project sets up a custom VPC containing one public subnet and one private subnet, and deploys an EC2 instance in each. It demonstrates how AWS networking components — the Internet Gateway, route tables, and security groups — work together to expose one instance to the internet while keeping the other isolated, and how connectivity between the two can still be verified over their private IP addresses.

Aim / Objective

To launch one Amazon EC2 instance in a Public Subnet and another Amazon EC2 instance in a Private Subnet within the same VPC, and to verify network connectivity between them using their private IP addresses.

Scope of the Project

- Design and provision a custom VPC (10.0.0.0/16) with segmented public and private address spaces.

- Configure Internet Gateway access for the public subnet only, keeping the private subnet unreachable from the internet.
- Deploy and configure EC2 instances with appropriate security group rules for SSH and ICMP traffic.
- Validate the design empirically through live SSH access and ping-based connectivity testing.

Tools & Technologies Used

- Amazon VPC — custom network, public & private subnets, route tables
- Amazon EC2 — Amazon Linux instances in each subnet
- Internet Gateway — public internet access for the public subnet
- Security Groups — instance-level firewall rules (SSH, ICMP)
- SSH / EC2 Instance Connect — remote access and connectivity testing
- AWS Management Console — provisioning and verification

Methodology

The work was carried out in four sequential phases, each owned by a group member:

- Phase 1 — Sohit Sharma: Created the custom VPC (10.0.0.0/16) and the public (10.0.1.0/24) and private (10.0.2.0/24) subnets.
- Phase 2 — Mudit Chaudhary: Attached the Internet Gateway, configured the public and private route tables, and set up the security groups for SSH and ICMP access.
- Phase 3 — Prince (Group Leader): Launched the Public and Private EC2 instances with the correct subnet, IP assignment, and security group settings.
- Phase 4 — Anshika Rai: Connected to the public instance over SSH and confirmed connectivity to the private instance using ping over its private IP address.

Expected Outcome

Both EC2 instances launch successfully in their respective subnets within the same VPC. The public instance is reachable from the internet via SSH, while the private instance remains inaccessible externally. Despite this isolation, the public instance is able to reach the private instance directly over its private IP address, confirming that intra-VPC routing and the configured security group rules work as intended.

Applications / Usage

This public–private subnet pattern is a foundational building block in real-world AWS architecture, and the skills practiced here carry directly into production scenarios such as:

- Three-tier web applications — public subnet hosts web/app servers; private subnet hosts the database, hidden from direct internet access.
- Bastion host architectures — a single hardened public instance is used as the only entry point into private resources, reducing the attack surface.

- Backend and internal services — application servers, caches, and internal APIs that should never be directly internet-reachable.
- A hands-on foundation for AWS networking concepts assessed in cloud certifications (e.g., AWS Certified Solutions Architect, AWS Certified Cloud Practitioner).

Limitations

- No NAT Gateway is configured, so the private instance has no outbound internet access at all (e.g., it cannot download OS updates or packages).
- The setup uses a single Availability Zone, so it is not resilient to an AZ-level outage.
- The public instance's SSH rule is scoped to "My IP"; it must be updated manually whenever that IP address changes.
- Only one instance is deployed per subnet — there is no load balancing or auto-scaling.
- Connectivity is validated only at the network layer (SSH login and ICMP ping); no application-layer testing is performed.
- All resources are provisioned manually through the AWS Console rather than as Infrastructure as Code, so the setup is not easily repeatable or version-controlled.

Future Scope

- Add a NAT Gateway to the private route table so the private instance can reach the internet outbound (e.g., for updates) without being reachable inbound.
- Extend the design across multiple Availability Zones for high availability.
- Introduce an Auto Scaling Group and an Application Load Balancer in front of the public subnet.
- Automate the entire setup using Infrastructure as Code (e.g., AWS CloudFormation or Terraform) for repeatability.
- Replace direct SSH bastion access with AWS Systems Manager Session Manager for keyless, auditable access.

Conclusion

This project demonstrates a practical, secure network design pattern used widely in production AWS environments: separating internet-facing and internal resources within a single VPC. By the end of the project, the group will have hands-on experience provisioning VPCs, subnets, route tables, security groups, and EC2 instances, and validating network connectivity between isolated resources — core skills for cloud and DevOps engineering.